Project Report: Adaptive RIO (A-RIO) Queue Management in ns-3

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CSE322 Project

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1 Objective

This project implements and evaluates Adaptive RIO (A-RIO) in ns-3, basically implementing the idea discussed in [1] with focused analysis on queue behavior, drops, throughput, delay, delivery ratio, and drop ratio under wired and wireless scenarios.

2 a) Network Topologies Under Simulation

Two topologies were used:

1) Wired Dumbbell Topology

- Multiple source nodes connect to Router-1 via access links.
- Router-1 to Router-2 is the bottleneck link.
- Router-2 connects to a sink node via a high-capacity egress link.
- A-RIO (or RIO/RED for comparison runs) is installed on the bottleneck egress queue at Router-1.

2) Wireless Access + Wired Bottleneck Topology

- Sources are Wi-Fi STA nodes.
- Router-1 acts as Wi-Fi AP.
- Router-1 to Router-2 remains a wired bottleneck.
- Router-2 to sink remains wired egress.
- This hybrid topology preserves bottleneck queue analysis while introducing wireless contention/loss and node-energy effects.

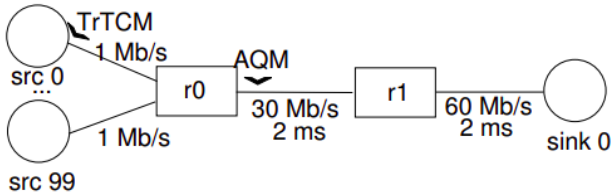


Figure 1: Dumbbell model

3 b) Parameters Under Variation

The following parameters were varied in experiments:

Table 1: Main Parameter Sweep

Parameter	Values
Number of nodes ($nNodes$)	20, 40, 60, 80, 100
Number of flows ($nFlows$)	10, 20, 30, 40, 50
Packets per second (pps)	100, 200, 300, 400, 500
Assured rate profile	25%, 50%, 75%, 100%, 125%
Traffic case	Case 1 to Case 6 (FTP/OnOff and RTT variants)

For wireless report simulations:

- Mobility speed: 5, 10, 15, 20, 25 m/s (when mobility enabled).
- Coverage side length: $k \times TxRange$, where $k = 1, 2, 3, 4, 5$ (for static-node studies).
- Energy model enabled for wireless nodes (STA + AP).

4 c) Modifications Made in the Simulator

The simulator was extended in the following ways:

Queueing and marking logic

- Implemented/updated A-RIO queue disc behavior in traffic-control module.
- Added trTCM-style packet coloring for green/yellow/red marking.
- Preserved RIO and RED versions for baseline comparisons.
- Original paper implements A-RIO with the same threshold for all the packets regardless of red, yellow, green. But here each type of packet was given a different threshold.

Experiment scripts

- Extended report simulations to accept run-time knobs: `nNodes`, `nFlows`, `pps`, and assured-rate profile.

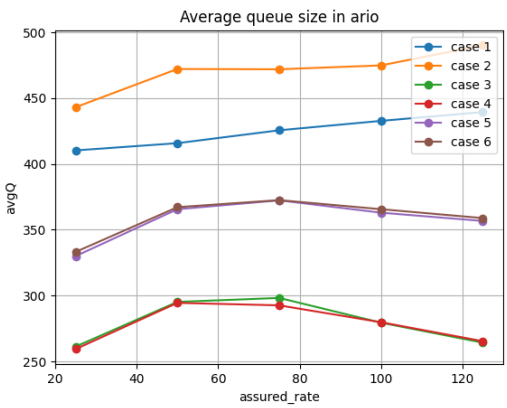
- For wireless reports, added knobs: `mobility`, `nodeSpeed`, `txRange`, `areaMultiplier`, and `initialEnergyJ`.
- Added FlowMonitor-based metric extraction:
  - network throughput,
  - average end-to-end delay,
  - packet delivery ratio (PDR),
  - packet drop ratio.
- Added wireless energy-consumption extraction from ns-3 energy sources.

Output and plotting

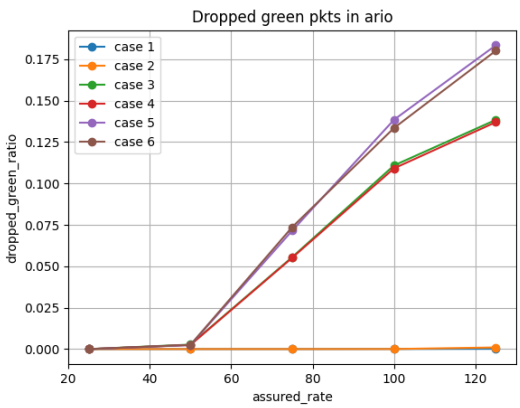
- CSV outputs were generated for report runs.
- Python plotting scripts generated all included figures in `scratch/`.

5 d) Results with Graphs

A-RIO assured-rate behavior (wired and wireless)

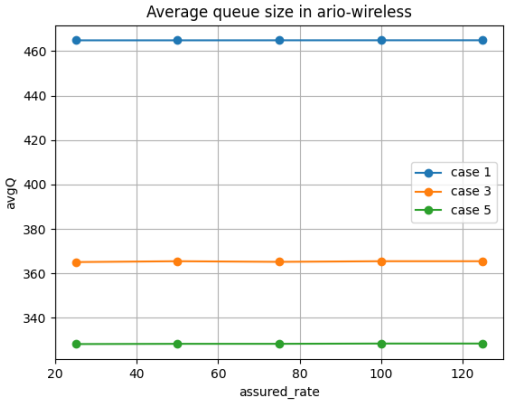


(a) A-RIO (wired): Avg queue vs assured rate

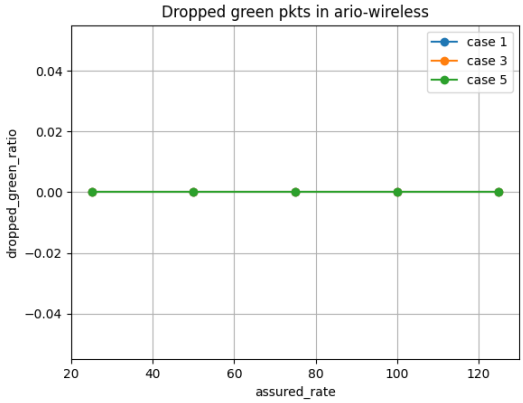


(b) A-RIO (wired): Green drops vs assured rate

Figure 2: A-RIO wired behavior with assured-rate variation



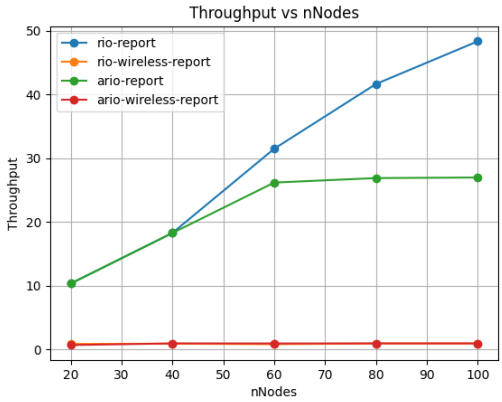
(a) A-RIO (wireless): Avg queue vs assured rate



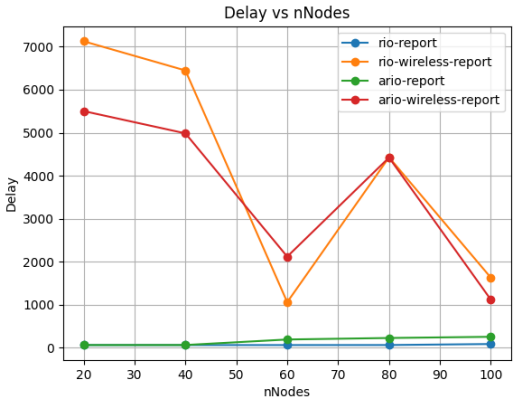
(b) A-RIO (wireless): Green drops vs assured rate

Figure 3: A-RIO wireless behavior with assured-rate variation

A-RIO report metrics vs node count

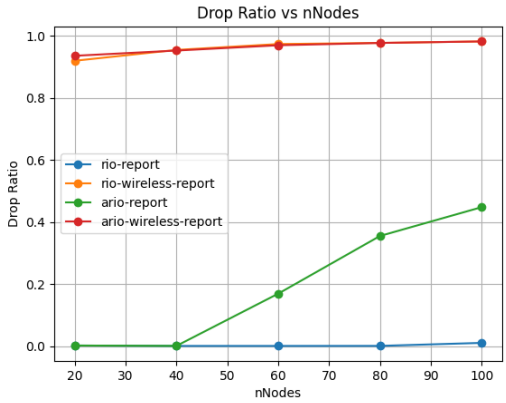


(a) Throughput vs nodes



(b) Avg delay vs nodes

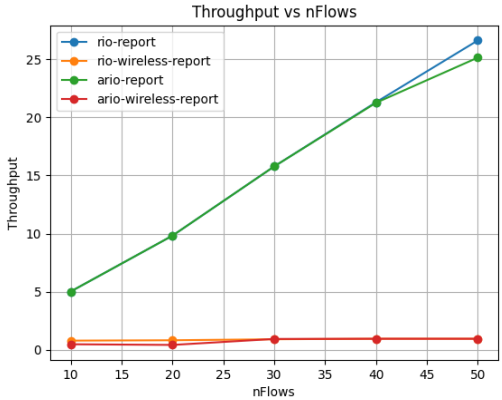
(c) Delivery ratio vs nodes



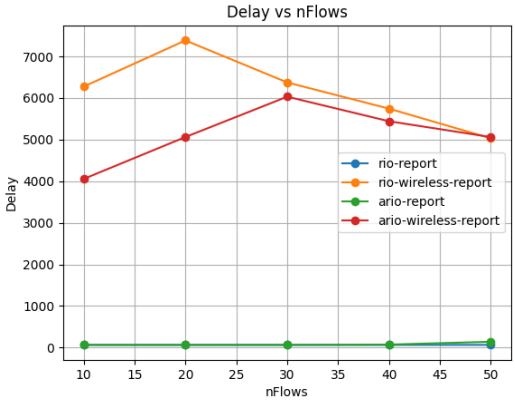
(d) Drop ratio vs nodes

Figure 4: A-RIO report metrics with varying node count

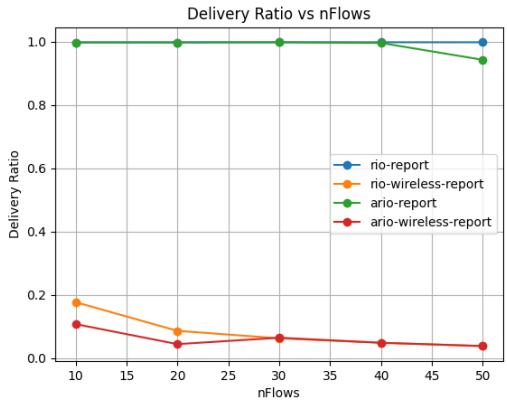
A-RIO report metrics vs flow count



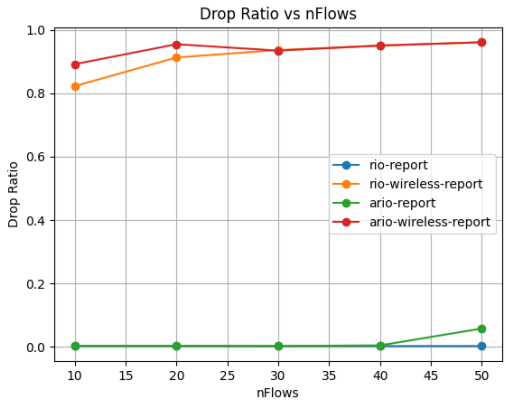
(a) Throughput vs flows



(b) Avg delay vs flows



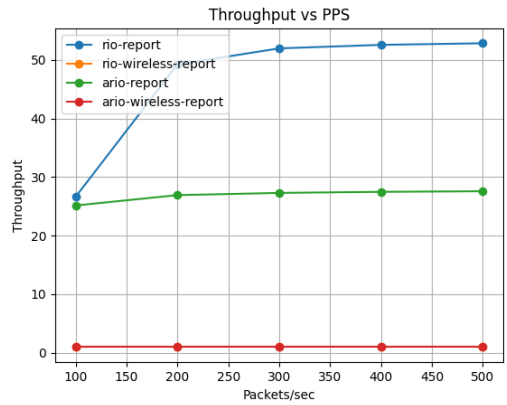
(c) Delivery ratio vs flows



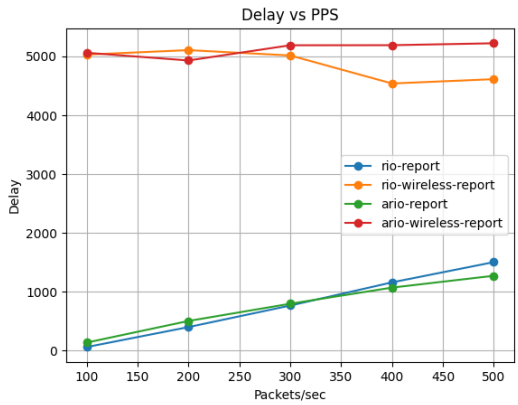
(d) Drop ratio vs flows

Figure 5: A-RIO report metrics with varying flow count

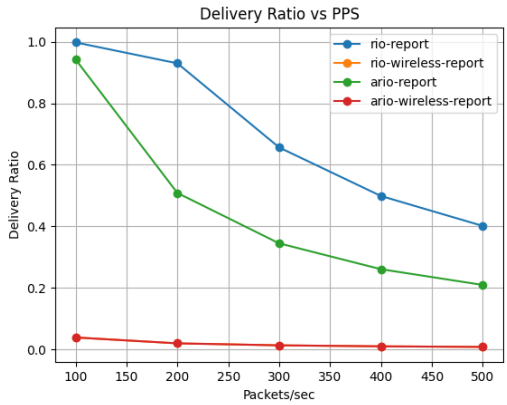
A-RIO report metrics vs offered packet rate



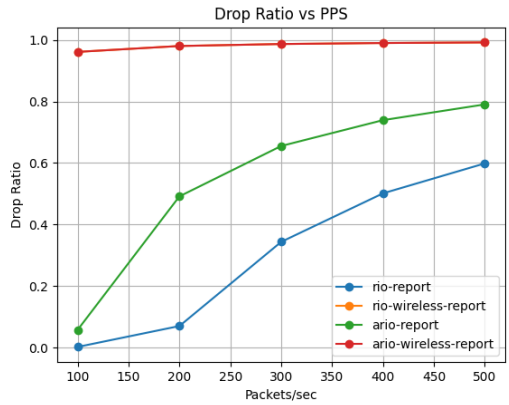
(a) Throughput vs pps



(b) Avg delay vs pps



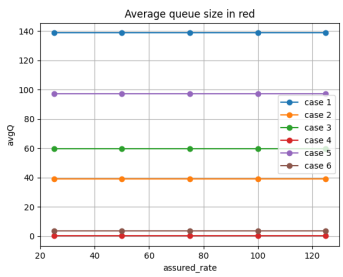
(c) Delivery ratio vs pps



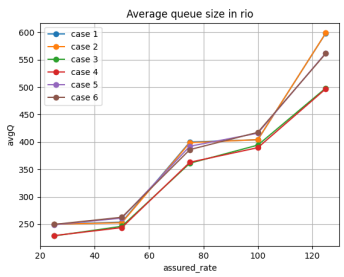
(d) Drop ratio vs pps

Figure 6: A-RIO report metrics with varying packet rate

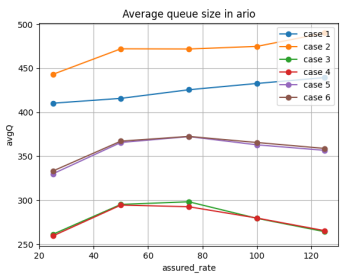
Baseline comparison (context)



(a) RED



(b) RIO



(c) A-RIO

Figure 7: Average queue comparison across RED, RIO, and A-RIO (wired)

6 e) Summary Findings

- A-RIO can maintain differentiated behavior under color-marked traffic while adapting drop tendency through dynamic control.

- As node count, flow count, and offered packet rate increase, throughput eventually saturates while delay and drop ratio increase.
- Delivery ratio generally decreases in high-contention regimes, especially in wireless scenarios.
- Wireless access introduces substantial additional loss and delay due to contention/channel effects, even when bottleneck queue drops are moderate.
- Energy consumption increases with greater offered load and channel activity.

7 f) Reflection on Findings (Discussion)

The most important observation is that end-to-end performance metrics and bottleneck-queue drop counters do not always move together. In several wireless runs, total drop ratio can be high while queue-disc drops remain relatively smaller, implying that many losses occur outside the bottleneck AQM (e.g., MAC/PHY contention or packets not delivered before simulation stop). This demonstrates why FlowMonitor-based end-to-end metrics are essential in addition to queue-level counters.

The parameter sweeps also show the need to tie sender behavior carefully to experiment knobs. If offered traffic does not change with a profile knob, queue and throughput curves may appear artificially flat. The report-oriented simulation versions improved this by exposing explicit runtime controls for nodes, flows, packet rate, mobility, and coverage.

Overall, A-RIO provided richer control than static-drop schemes in these scenarios, but wireless behavior highlighted that queue management alone is not enough to guarantee high delivery at heavy load. Future work can include per-flow fairness analysis, confidence intervals over multiple seeds, and direct side-by-side A-RIO vs RIO statistical significance testing.

References

[1] Julio Orozco and David Ros. "An Adaptive RIO (A-RIO) Queue Management Algorithm". In: *Lecture Notes in Computer Science*. Vol. 2811. 2003, pp. 11–20. DOI: 10.1007/978-3-540-45188-4_2.